EXP 9: Orchestrating Serverless Functions with AWS Step Functions

Theory:

Serverless functions let developers code in any language or framework with which they're comfortable. Simpler backend code. Serverless removes a lot of coding complexity for developers, allowing them to create simple, self-contained functions that independently perform one purpose.

Steps to Execute:

Step1: Create Step Function (State Machine)

- Go to **AWS Step Functions Console**.
- Click **Create State Machine**.
- Choose **Create your own**.
- Enter name: course_selection_state_machine

Step 2: Define the State Machine

- Paste the following JSON code:

```
{
  "Comment": "Student selection based on subject and marks",
  "StartAt": "StartHere",
  "States": {
    "StartHere": {
      "Type": "Pass",
      "Next": "SubjectChoice"
    },
    "SubjectChoice": {
      "Type": "Choice",
      "Choices": [
        {
          "Variable": "$.Subject",
          "StringEquals": "Physics",
          "Next": "Physics"
        },
        {
          "Variable": "$.Subject",
          "StringEquals": "Maths",
          "Next": "Maths"
        }
      ],
      "Default": "EndState"
    },
    "Physics": {
      "Type": "Pass",
      "Next": "CheckMarks"
    },
    "Maths": {
      "Type": "Pass",
      "Next": "CheckMarks"
    },
```

```
    "CheckMarks": {
      "Type": "Choice",
      "Choices": [
        {
          "Variable": "$.Marks",
          "NumericGreaterThan": 70,
          "Next": "EndState"
        }
      ],
      "Default": "EndState"
    },
    "EndState": {
      "Type": "Pass",
      "End": true
    }
  }
}
```

- Click **Next / Create State Machine**.

Step 3: Execute the State Machine

- Click **Start Execution**.
- Provide input JSON (example):

```
{
  "Subject": "Maths",
  "Marks": 76
}
```

OR

```
{
  "Subject": "Physics",
  "Marks": 91
}
```

- Run the execution.

Step 4: Observe Output

- View the **Graph View**.
- Observe each step turning **green (successful execution)**.
- Check how the flow moves based on subject and marks.

Step 5: Cleanup

- Delete the created **State Machine** after completion.

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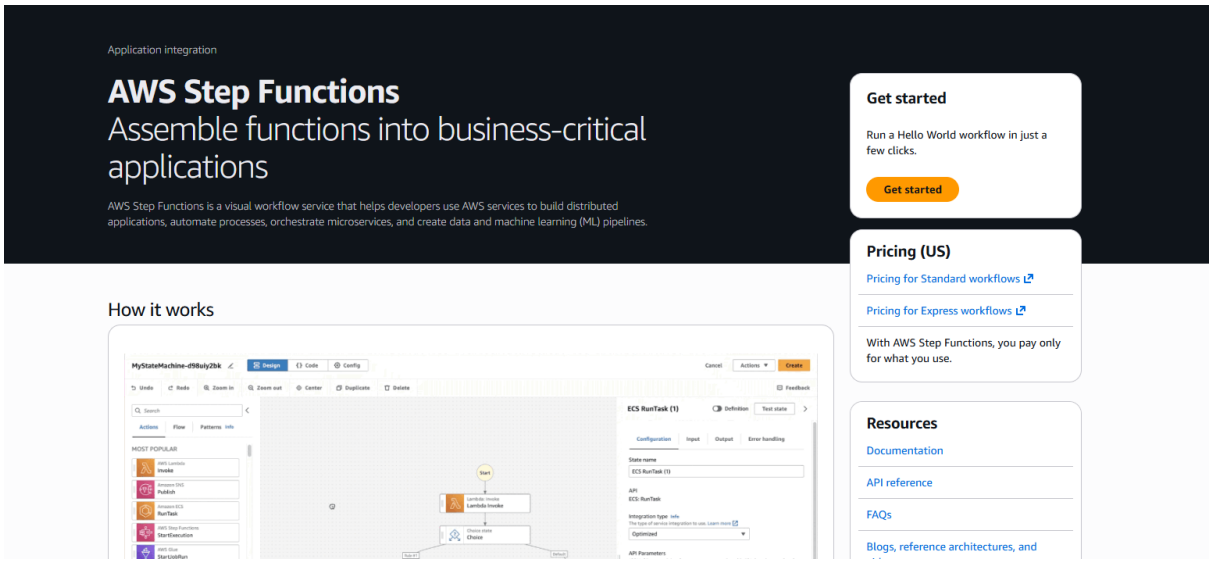
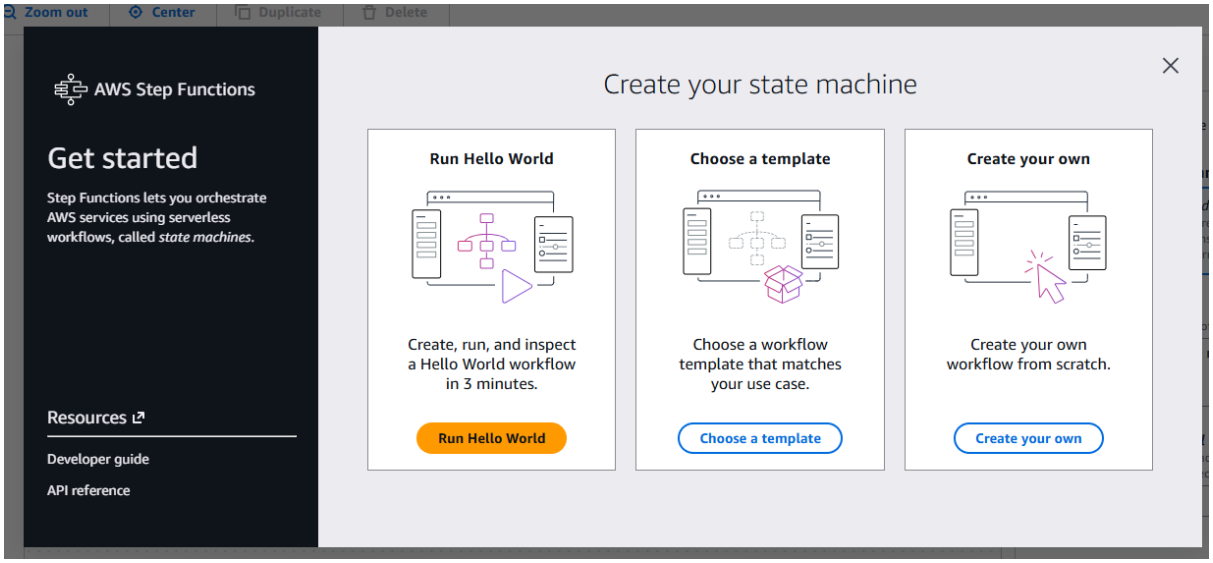
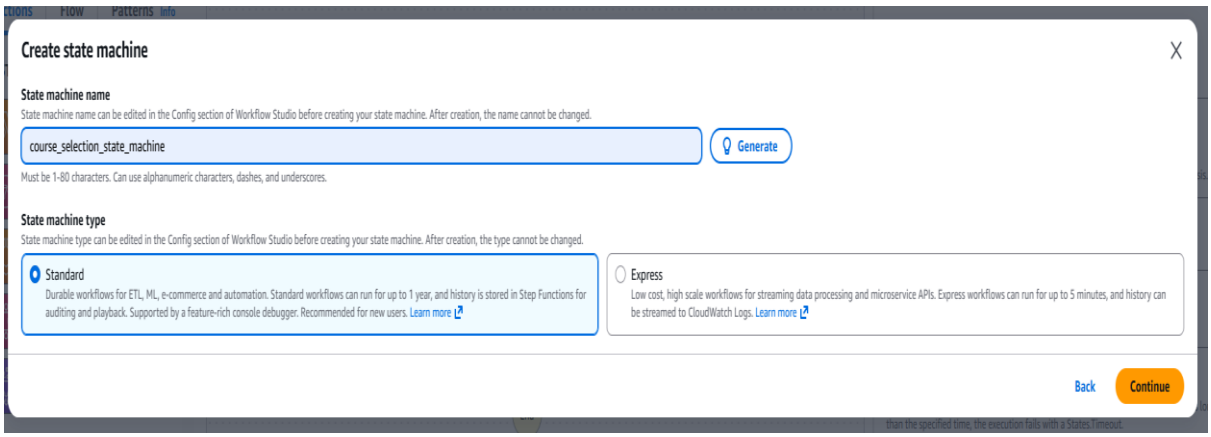






course_selection_state_machine

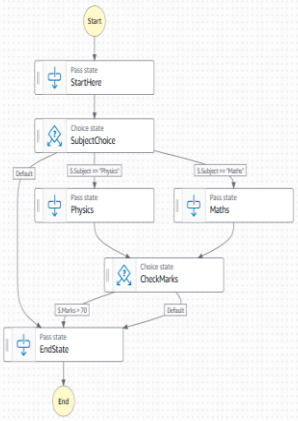
Design | Code | Config

Cancel | Actions

Undo | Redo | Format | Copy | Commands | View docs

```
        "StringEquals": "Physics",
        "Next": "Physics"
      },
      {
        "Variable": "$.Subject",
        "StringEquals": "Maths",
        "Next": "Maths"
      }
    ],
    "Default": "EndState"
  },
  "Physics": {
    "Type": "Pass",
    "Next": "CheckMarks"
  },
  "Maths": {
    "Type": "Pass",
    "Next": "CheckMarks"
  },
  "CheckMarks": {
    "Type": "Choice",
    "Choices": [
      {
        "Variable": "$.Marks",
        "NumericGreaterThan": 70,
        "Next": "EndState"
      }
    ],
    "Default": "EndState"
  },
  "EndState": {
    "Type": "Pass",
    "End": true
  }
}
}
```

Zoom in | Zoom out | Center



Confirm role creation

An execution role will be created with full permissions.
A new execution role named *StepFunctions-course_selection_state_machine-role-8sr7bphrj* will be created. All required permissions for the actions specified in your state machine will be auto-generated.

Role permissions

Service	Action(s)	Status	Documentation links
AWS X-Ray	xray:PutTraceSegments xray:PutTelemetryRecords xray:GetSamplingRules xray:GetSamplingTargets	Policies will be generated for X-Ray tracing	X-Ray policies for Step Functions

Cancel | View role configuration | Confirm

course_selection_state_machine

Edit | Actions | Start execution

Details

Arn
arn:aws:states:eu-north-1:816069141836:stateMachine:course_selection_state_machine

IAM role ARN
arn:aws:iam::816069141836:role/service-role/StepFunctions-course_selection_state_machine-role-8sr7bphrj

Type
Standard

Status
Active

Creation date
Apr 17, 2026, 12:06:22 (UTC+05:30)

X-Ray tracing
Disabled

Executions | Monitoring | Logging | Definition | Aliases | Versions | Tags

Executions (0/0)

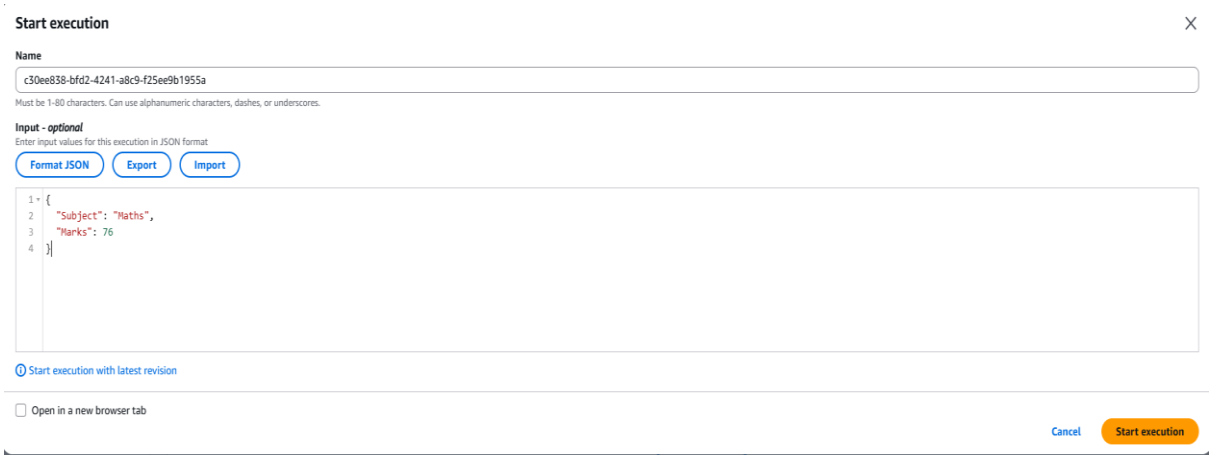
View details | Stop execution | Redrive | Start execution

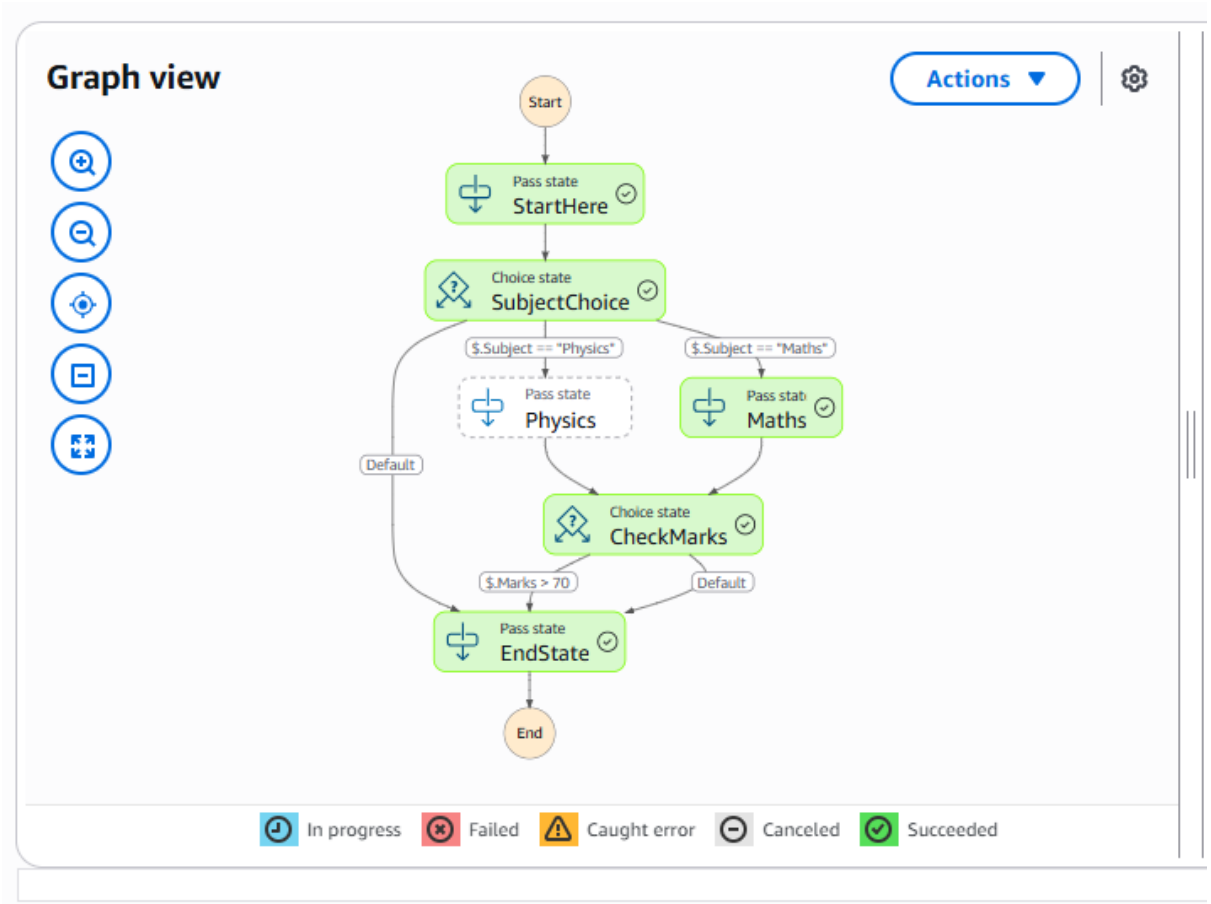
Filter executions by property or value | All | Last 15 months | Local timezone | 0 matches | 1

Name	Status	Start Time (local)	End Time (local)	Duration Info	Version	Alias

No executions

Start execution


Start execution
Name
c30ee838-bfd2-4241-a8c9-f25ee9b1955a
Must be 1-80 characters. Can use alphanumeric characters, dashes, or underscores.
Input - optional
Enter input values for this execution in JSON format
Format JSON
Export
Import
{
"Subject": "Maths",
"Marks": 76
}
Start execution with latest revision
Open in a new browser tab
Cancel
Start execution


Graph view
Actions
Start
Pass state
StartHere
Choice state
SubjectChoice
$.Subject == "Physics"
$.Subject == "Maths"
Pass state
Physics
Pass state
Maths
Default
Choice state
CheckMarks
$.Marks > 70
Default
Pass state
EndState
End
In progress
Failed
Caught error
Canceled
Succeeded